

Virtual Try-On AI

Project Workflow & Research Notes

Module 1 | Module 2 | Module 3 | Module 4

Body Detection → Pose Estimation → Cloth Segmentation → Try-On

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Module 1 — Body Detection System

Body Detection is the very first step in the whole Virtual Try-On pipeline. Before the system can do anything — like estimating pose or overlaying a garment — it first needs to know where the person is in the image. That is what body detection does.

What it does

It detects the human body in the photo and draws a bounding box around the person. This tells the system — okay, the person is here, now focus only on this region. It also separates the person from the background so further processing is clean and accurate.

How it works

- Uses object detection models like **YOLO** or **MediaPipe** to find the person
- Draws a bounding box around the detected human
- Passes that cropped region to the next module

Workflow

Input Image → Detection Model → Bounding Box → Cropped Person Region

Tools / Models Used

Model	Purpose
YOLO v8	Fast real-time body detection
MediaPipe	Lightweight, works on mobile too
OpenCV	Image processing and bounding box drawing

Why it matters

Without body detection the system doesn't know where the person is. It might try to apply clothing on the background or some random region. So this module is the foundation — get the person location right, everything else follows properly.

Module 2 — Human Pose Estimation

Pose Estimation is about understanding how the person is standing or sitting — what position their body is in. It detects specific keypoints on the body like shoulders, elbows, wrists, hips and so on. These keypoints are then used to figure out how to place the garment correctly.

What it does

- Detects **17–33 body keypoints** from the person photo
- Maps out where shoulders, chest, waist, arms are
- Tracks **skeletal structure** and body movement
- Passes keypoint data to garment warping module

How it works

The model looks at the image and predicts heatmaps for each body part. Where the heatmap is highest — that is where the keypoint is. For example the shoulder heatmap will peak at the exact shoulder position. From all these keypoints together the full body pose is understood.

Workflow

```
Person Image → Pose Model → Keypoint Detection → Skeleton Map → Body Measurements
```

Tools Used

Tool	Purpose
MediaPipe Pose	Real-time 33 keypoint detection
OpenPose	Research grade, more accurate
HRNet	High resolution keypoint detection

Task done in this module

- Understood the pose estimation workflow
- Studied skeletal tracking and body movement
- Took screenshots of pose keypoint outputs
- Experimented with MediaPipe Selfie Segmentation

Module 3 — Cloth Segmentation AI

(How AI Understands Clothing)

Cloth Segmentation is how the AI understands and separates the clothing in an image. It identifies all the different regions — like where the shirt is, where the pants are, where the skin is — and separates them out. This is needed so the system knows exactly which part to replace with the new garment.

What it does

- Separates old clothes from the body region
- Identifies all clothing regions in the image
- Creates a **mask** of the clothing area
- Highlighted region is pre-processed for warping

How it works

The segmentation model looks at every pixel in the image and classifies it — is this pixel part of shirt, pants, skin, background or something else? This is called pixel-wise classification. The output is a colored mask where each color represents a different region.

Workflow

```
Input → AI Seg Model → Identify Clothing Region → Mask (Separated) → Highlighted Region
```

Key Concepts

- **Segmentation Models** — AI models trained to classify regions in an image
- **DeepFashion Dataset** — Used for training fashion AI models. Contains thousands of labeled clothing images
- **Mask Output** — Binary or colored map showing where the garment is

Tasks done

- Understood the Segmentation Workflow
- Tested the segmentation model
- Studied the DeepFashion Dataset
- Experimented with MediaPipe Selfie Segmentation

Note

The segmentation at this stage does not have 100% accuracy — there can be small errors at the edges. That is handled and refined in the garment warping stage later.

Module 4 — Virtual Try-On AI

Virtual Try-On (VTO) is the final module where everything comes together. It uses AI to simulate how a garment looks on a person without them physically wearing it. You upload a person photo and a garment image — the system shows you the result as if you are wearing it.

Garment Warping

Garment warping takes the flat garment image and adjusts it to fit the person's body shape. It stretches, bends and wraps the cloth around the detected body keypoints so the output looks like the person is actually wearing it and not just a sticker pasted on top.

- Stretches or adjusts garment to fit different body shapes
- Uses **TPS (Thin Plate Spline)** technique — like rubber sheet warping
- Gives a realistic natural look to the garment

AI Fitting System

The fitting system figures out whether the garment actually fits the person. It estimates body measurements from the photo and compares them with the garment's size to give a fit result — S, M, L or XL. It also checks if the cloth sits correctly on the body.

- Estimates fitting from body keypoints
- Checks shoulder width, chest area, torso length
- Simulates the clothing in an app environment

Complete Pipeline

Person Photo → Background Removal → Pose Estimation → Cloth Segmentation
→ Garment Warping → Overlay → Try-On Output

Task Breakdown

Task	Status
Understand the Workflow (W/F)	Done
Study Pipelines	Done
Connect previous Modules	Done
Overlay Shirt Image (Experiment)	Done
Test the output	Done
Study Demo / Create Demo	Done
Create Simple Mock Workflow	Done

Research — Garment Warping & Fitting System

Garment Warping

So basically garment warping is how AI takes a flat cloth image and makes it look like it is actually worn on a person. If you just paste a shirt on a photo it looks weird and fake — that is where warping comes in and fixes that.

It detects the body shape first — shoulders, chest, waist — and then bends and stretches the cloth to fit around those points. The main technique used here is called TPS (Thin Plate Spline). Think of it like a rubber sheet — you pull the corners and the whole sheet adjusts smoothly in between.

Steps in Garment Warping

- Take the flat garment image
- Detect body keypoints from the person photo
- Warp the garment to match those keypoints
- Output looks natural — like person is wearing it

The models mostly used for this are **CP-VTON** and **HR-VITON**. They are trained on thousands of person and garment pairs so the warping output looks realistic and not artificial.

AI Fitting System

AI Fitting System is how the system figures out the right fit for a garment on a person just by looking at their photo. No measuring tape, no manual input — just the image and AI does the rest.

It uses Pose Estimation models like MediaPipe or OpenPose to detect key body landmarks from the photo. From those landmarks it calculates rough body measurements — shoulder width, chest area, torso length, hip region — and compares them with the garment's size chart.

What it checks

- Is the shirt too tight or too loose
- Are the shoulders aligning properly with the garment
- Is the length correct for the body
- Does the garment sit naturally on the body

Models Used

Model	Role
MediaPipe Pose	Lightweight real-time landmark detection
OpenPose	Detailed body point mapping for research
CP-VTON	Main cloth + person try-on model
HR-VITON	High resolution warping and fitting

TPS Warping	Geometric transformation of garment
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Why Both Together Matter

Warping makes it look real. Fitting system makes it size accurate. Without warping the overlay looks flat and fake. Without the fitting system it might show a wrong size. Together they make the full virtual try-on experience work properly — just like trying on a real garment in a store.

Submit Checklist & Workflow Summary

Submit Checklist

#	Deliverable	Status
1	Simple Alignment Demo (Drag & Drop Try-On App)	Done
2	Workflow Notes (All 4 Modules)	Done
3	Research — Garment Warping	Done
4	Research — AI Fitting System	Done
5	Screenshots (S.S) of Background Removal	Pending
6	Screenshots (S.S) of Try-On Output	Pending

Full Project Workflow Summary

This project is about building a complete Virtual Try-On AI pipeline from scratch by connecting four modules together. Each module handles one specific part of the problem and passes its output to the next.

Module	Name	What it does
M-1	Body Detection	Detect person in image, draw bounding box
M-2	Pose Estimation	Detect 17-33 body keypoints, skeletal map
M-3	Cloth Segmentation	Identify and mask clothing regions
M-4	Virtual Try-On	Warp garment + overlay + fitting check

References to Study

- **VITON** — Original Virtual Try-On research paper
- **HR-VITON** — High Resolution Virtual Try-On model
- **CP-VTON** — Cloth + Person Try-On model
- **MediaPipe Pose** — Google's body landmark detection
- **DeepFashion Dataset** — Fashion AI training dataset